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Assignment Set – 1

Q1(a). Discuss the objectives of Time Series Analysis. [5 Marks]

Time Series Analysis is a way to look at data that is collected over time. It helps us understand what is happening. Make good guesses about what will happen in the future. Time Series Analysis is very important in areas like money, health getting things from one place to another and weather.

The main goals of Time Series Analysis are as follows:

1. **Description:** The first goal is to say what is happening with the data over time. This means looking at trends what happens at the time every year and things that do not happen regularly. We use numbers and pictures to understand what is going on.
2. **Explanation:** Time Series Analysis helps us understand why things are changing. We break down the data into parts like what's happening over time what happens at the same time every year things that happen in cycles and things that are not regular. This helps us see what is causing the changes.
3. **Forecasting:** This is an important goal. We use data to guess what will happen in the future. We use models like ARIMA, Exponential Smoothing and Prophet to make guesses about what will happen soon and what will happen later. This is very important for planning and making sure we have things.
4. **Control:** Time Series Analysis helps us watch and control things. In making things and checking quality we use pictures to see if things are not going as planned. This means we can fix problems quickly.
5. **Intervention Analysis:** This goal is about seeing how certain events or changes affect the data. For example we can see how a new government rule affects stock prices or how a marketing plan affects sales.
6. **Decomposition:** We break down the data into parts like what's happening over time what happens at the same time every year things that happen in cycles and things that are left over. This helps us see what is causing what.
7. **Hypothesis Testing:** Time Series Analysis helps us test ideas about the data. We can see if a trend is real or if two things are connected.

In short Time Series Analysis helps us understand data that changes over time. This helps organizations make decisions guess what will happen next and control things that are always changing. Time Series Analysis is very useful, for making guesses about the future. Time Series Analysis is used in areas to help us understand what is happening with the data over time.

Q1(b). Explain Autoregressive Model. [5 Marks]

An Autoregressive model is a type of model that helps us guess what will happen next in a series of events. It does this by looking at what happened and using that information to make a prediction. The basic idea is that what is happening now can be explained by what happened plus some random things that we cannot predict.

The Autoregressive model is one of the important models we use to understand and predict things that happen over time.

Mathematical Representation:

An model of order p , which we call $AR(p)$ can be written like this:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

Where:

- Y_t is what is happening now in the series
- c is a number
- $\phi_1, \phi_2 \dots \phi_p$ are the Autoregressive coefficients that help us make predictions
- $Y_{t-1} Y_{t-2} \dots Y_{t-p}$ are what happened before in the series
- ε_t is the random part that we cannot predict
- p is the order of the Autoregressive model, which tells us how many things that happened before we should look at

Key things to know about the Autoregressive model:

1. **The series needs to be stable:** For the Autoregressive model to work well the series of events needs to be stable meaning that what happens on average and how much things vary does not change over time. If the series is not stable we need to adjust it before using the Autoregressive model.
2. **Choosing the order:** We use special tools like the Partial Autocorrelation Function to decide how many things that happened before we should look at. This helps us find the order for the Autoregressive model.
3. **Model of order 1:** This is the simplest case, where what happens now depends only on what happened right before. We often use this to understand things and stock prices.

4. **Model of order 2:** This is a bit more complicated, where what happens now depends on the two things that happened before. This helps us understand complex patterns over time.

We use the model for many things like:

- Guessing what will happen in financial markets, such as stock prices and exchange rates
- Predicting economic indicators like how much money a country has and how much things cost
- Understanding the weather and climate
- Processing signals

The Autoregressive model is very important because it is the basis for more complex models, like ARMA and ARIMA which we use to understand and predict things that happen over time.

Q2(a). Interpret the ETS Model. [5 Marks]

The ETS model is an useful tool for predicting what will happen in the future. It is also known as the Exponential Smoothing State Space Model. This model helps us understand three parts of a time series: error, trend and seasonality. It gives us a way to use statistics to make predictions.

Components of the ETS Model:

1. **Error:** This is the part that we cannot explain with trend or seasonality. It can be one of two types:

- **Additive:** The error is always the same no matter what is happening in the series.
- **Multiplicative:** The error changes depending on what's happening in the series.

2. **Trend:** This is the direction that the data is moving in the term. It can be:

- **None:** There is no trend.
- **Additive:** The series is changing by the amount each time.
- **Additive Damped:** The trend is getting weaker over time.
- **Multiplicative:** The trend is changing depending on what's happening in the series.

3. **Seasonality:** This is when we see patterns that happen at the time every year. It can be:

- **None:** There is no pattern.
- **Additive:** The pattern is always the same.
- **Multiplicative:** The pattern changes depending on what's happening in the series.

We use a way of writing the ETS model like this: ETS(E, T, S). For example:

- **ETS(A, A A)** means we have error, additive trend and additive seasonality. This is the same as the Holt-Winters method.
- **ETS(M, A, M)** means we have multiplicative error, additive trend and multiplicative seasonality.
- **ETS(A, N N)** means we have additive errors, no trend and no seasonality. This is the same as exponential smoothing.

The ETS model works by using two equations:

- **The measurement equation:** This connects what we can see to what we cannot see.
- **The transition equations:** These show us how things change over time.

The ETS model has a lot of advantages:

- It gives us 30 models to choose from.
- It helps us choose the best model automatically.
- It gives us an idea of what will happen in the future.
- It works well with different types of time series.

The ETS model is really good at predicting what will happen in the term and medium term. It is useful for people who work in retail, inventory planning and finance. The ETS model is a tool, for predicting what will happen in the future and it is widely used. The ETS model helps us make predictions and it is an important part of time series forecasting.

Q2(b). Define the ARCH Model. Explain its usage. [5 Marks]

Definition: The ARCH model, introduced by Robert Engle in 1982 is a model. It helps us understand and model the changing volatility in data. Unlike models the ARCH model knows that the variance of errors can change over time. This change depends on the size of errors.

Mathematical Formulation:

An ARCH(q) model is defined as:

$$* Y_t = \mu + \epsilon_t$$

$$* \epsilon_t = \sigma_t \cdot z_t \quad z_t \sim N(0,1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2 + \dots + \alpha_q \epsilon_{t-q}^2$$

Where:

* σ_t^2 is the variance at time t

* $\alpha_0 > 0$ is a constant

* $\alpha_1, \alpha_2, \dots, \alpha_q \geq 0$ are parameters. They show how past squared errors affect the model.

* q is the order of the ARCH model

Key Features:

* The ARCH model helps us see that big changes are often followed by changes. Small changes are often followed by changes.

* The conditional variance depends on squared residuals. This makes it dynamic.

* It assumes that the long-term variance is constant. However the short-term variance changes.

Applications:

1. **Financial Risk Management:** We use models to estimate Value at Risk (VaR) and Expected Shortfall. These are metrics for measuring financial risk.

2. **Option Pricing:** models help us price options and derivatives more accurately. Volatility is an input here.

3. **Portfolio Optimisation:** By modelling changing volatility ARCH helps us allocate assets dynamically. We adjust weights based on changing risk profiles.

4. **Stock Market Analysis:** The ARCH model captures volatility patterns in stock returns, bond yields and exchange rates. It provides realistic models than those assuming constant variance.

5. **Forecasting Volatility:** The ARCH model helps us forecast volatility. This is essential, for trading strategies, hedging decisions and monetary policy analysis.

The ARCH model led to the GARCH (Generalised ARCH) model. The GARCH model is widely used in practice.

Q3(a). Describe a few risks associated with Artificial Intelligence. [5 Marks]

Artificial Intelligence or AI has the power to change things a lot. It also has some big problems that people and organisations need to think about carefully.

1. Bias and Discrimination:

AI systems learn from data and if that data is not fair. Like it is biased against some people because of their race or where they come from. Then the AI will do the same thing. For example some tools that use AI to help hire people have been found to be unfair to some groups because the data they learned from was not fair. This is a problem, especially in important areas like the police, banks and hospitals.

2. Lack of Transparency and Explainability:

Some AI systems, like the ones that use lots of layers are like boxes. You cannot see how they make decisions. This makes it hard for people to understand why the AI did something. It is a problem when we need to know why. This lack of understanding makes it hard to trust the AI. It is also a problem when we need to follow rules especially in areas like money and healthcare where we need to be accountable.

3. Job Displacement and Economic Disruption:

AI is taking over some jobs, which's a big problem for people who work. AI can do some tasks, like the ones that're the same every day much faster and better than people. While AI is making jobs it is hard for people who lose their jobs to get new ones because they do not have the skills.

4. Security and Privacy Threats:

AI systems can be used to do things like make fake videos or hack into computers. Some people can also trick the AI into making mistakes, which's a big problem. Also AI needs a lot of data to work. This is a problem when it comes to peoples private information.

5. Autonomous Decision-Making Risks:

As AI systems become more independent, like self-driving cars there is a risk that they will make mistakes that can hurt people. If an AI system makes a mistake it is not clear who is responsible which is a problem.

6. Concentration of Power:

a few big companies and countries are making AI, which means they have a lot of power. This is not fair. It can make it harder for other people and companies to catch up.

These problems show that we need to be careful when we make and use AI. We need to have rules and guidelines to make sure that AI is used in a way. Artificial Intelligence needs to be used.

Q3(b). Appraise some challenges or limitations we face with Deep Learning. [5 Marks]

Deep learning is a part of machine learning that uses neural networks with many layers. It has done some great things but there are still some big problems that need to be fixed.

1. Need for a Lot of Data:

Deep learning needs a lot of data to work well. This data needs to be labelled so the model can learn from it. In some areas like medical conditions it is hard to get this data. Collecting and preparing this data is also very expensive and time consuming.

2. High Cost of Computing:

Training neural networks needs a lot of computer power. This includes graphics cards, a lot of memory and a lot of electricity. This makes deep learning too expensive for medium sized businesses. Training one big language model can cost a lot of money and hurt the environment.

3. Not Being Able to Understand:

Deep learning models are like boxes. We cannot see how they make their decisions. This is a problem in areas like medical diagnosis, self driving cars and financial regulation. People need to understand how these models work so they can trust them.

4. Overfitting:

Deep networks can remember the data they are trained on of learning from it. This means they work well on the data they are trained on but not on data. There are some ways to stop this, like dropout and regularisation. It is still a problem.

5. Being Vulnerable to Attacks:

Deep learning models can be tricked into making decisions. This can be done by changing the input data in a way that's hard to notice. This is a problem for things like self driving cars and facial recognition systems.

6. Limits of Transfer Learning:

Transfer learning is when a model is trained on one task and then used on another task. This can work well. It does not work well if the two tasks are very different. This is something that people are still working on.

7. Gradients That Are Small or Too Big:

When training deep networks the gradients can become too small or too big. This can cause problems when training the model. Some types of networks, like LSTMs and ResNets have helped with this. It is still a problem.

In conclusion deep learning is still a powerful tool but we need to fix these problems to make it work better. We need to make algorithms use less power and think about the ethics of deep learning. This will help learning to keep getting better.

Assignment Set – 2

Q4(a). Discuss ANN classification models. [5 Marks]

Artificial Neural Network (ANN) models are a type of computer program that tries to mimic how our brains work. They are made up of connected parts (called neurons) that are organised in layers. These models are really good at helping us put data into categories.

Architecture of ANN

An ANN model typically has three parts:

1. **Input Layer:** This part gets the data. Each neuron here corresponds to one piece of data. For example if we are trying to guess if a customer will leave a company the data might

include their age how long they've been a customer, how much they pay each month and what kind of contract they have.

2. **Hidden Layers:** These are the parts where the model does most of its learning. Each neuron here takes the data adds some numbers together and then uses a rule (like ReLU, sigmoid or tanh) to decide what to do next. Having many of these layers helps the model understand relationships.

3. **Output Layer:** This part gives the answer. If we are only trying to choose between two things one neuron with a rule can give us a probability (0 or 1). If we have choices we use another rule that gives us a list of probabilities.

Training Process

1. **Forward Propagation:** The data goes through the model one layer at a time and the model makes a guess.

2. **Loss Computation:** The model checks how wrong its guess was using a formula (like binary cross-entropy for two choices or categorical cross-entropy for many choices).

3. **Backpropagation:** The model figures out how to change its numbers to make guesses next time and updates them using methods like Stochastic Gradient Descent (SGD) Adam or RMSprop.

4. **Iteration:** This process is repeated times until the model gets really good at making guesses.

Types of ANN Models

- **Perceptron:** This is the ANN with just one layer. It is good for data that can be separated easily.

- **MultiLayer Perceptron (MLP):** This has one or more middle layers and can understand complicated relationships.

- **Radial Basis Function (RBF) Networks:** These use special functions to help recognise patterns.

Applications

- Image recognition

- Sentiment analysis

- Medical diagnosis

- Fraud detection

- Customer segmentation

ANN models are powerful tools. When we set them up and train them properly they can be very accurate, at solving problems.

Q4(b). Explain the classification of layers of CNN. [5 Marks]

Convolutional Neural Networks are kinds of deep learning models. They are mainly used for processing images and other kinds of grid data.

1. Convolutional Layer:

The convolutional layer is the part of a Convolutional Neural Network. It uses filters that move across the input image. These filters do element- multiplication and summation to produce feature maps. Each filter finds features like edges, textures or patterns. Important things to consider are:

- Filter size, for example 3×3 or 5×5
- Stride which's the step size of filter movement
- Padding, which is zero-padding to keep spatial dimensions

Multiple convolutional layers together help the network learn different levels of features. Early layers learn low-level features like edges. Deeper layers learn high-level features like objects.

2. Activation Layer (ReLU):

After each convolution a special function called Rectified Linear Unit or ReLU is applied. ReLU changes all values to zero. This helps the model learn things.

3. Pooling Layer:

The pooling layer makes the feature maps smaller. This reduces computation. Helps prevent overfitting. There are two types:

- **Max Pooling:** It picks the biggest value in each window. This keeps the important features.

-**Average Pooling:** It calculates the average value in each window. This gives a downsampling.

Pooling helps the model understand the input better even if things move a little.

4. Connected Layer:

After convolutional and pooling layers the feature maps are changed into a one-dimensional vector. This vector is then fed into connected layers. These layers work like a neural network.

They combine features to make the classification. Each neuron in a connected layer is connected to every neuron in the previous layer.

5. Dropout Layer:

Dropout is a technique to prevent overfitting. It randomly ignores some neurons during training. A typical dropout rate is between 25% to 50%. This means 25% to 50% of neurons are not used in each training iteration.

6. Output Layer:

The final layer uses a function for classification. For classes it uses softmax. For two classes it uses sigmoid. This layer gives probability scores for each class.

These layers work together to help Convolutional Neural Networks learn features automatically. This makes them very good, for image classification, object detection and visual recognition tasks..

Q5(a). Describe the classification of RNN based upon architecture. [5 Marks]

Recurrent Neural Networks are a type of network that can handle data that comes in a sequence. Recurrent Neural Networks can remember things from the past, which helps them make decisions. We can group Recurrent Neural Networks into types based on how they handle input and output sequences. Here are some of the types of Recurrent Neural Networks:

1. One-to-One:

This is the type of Recurrent Neural Network. It has one input and one output. This type of network is not really a Recurrent Neural Network because it does not use the memory feature. We include it here just to be complete.

- **Example:** We can use this type of network to classify images.

2. One-to-Many:

This type of Recurrent Neural Network takes one input. Produces many outputs. The network gets one input. Then produces many outputs over time.

- **Example:** We can use this type of network to write a caption for an image. We can also use it to generate music from a note.

3. Many-to-One:

This type of Recurrent Neural Network takes inputs and produces one output. The network looks at all the inputs. Then produces one output at the end.

- **Example:** We can use this type of network to figure out if a review's positive or negative.

4. Many-to-Many:

This type of Recurrent Neural Network takes inputs and produces many outputs. There are two types of Many-to-Many networks.

- One type produces outputs that're the same length as the inputs.

- **Example:** We can use this type of network to identify the names of people and places in a sentence.

5. Many-to-Many:

This type of Recurrent Neural Network takes inputs and produces many outputs but the outputs are not the same length as the inputs. This type of network is called an encoder-decoder network. The encoder part of the network looks at the inputs. Then the decoder part produces the outputs.

- **Example:** We can use this type of network to translate a sentence from one language to another. We can also use it to summarize a piece of text.

There are also some advanced types of Recurrent Neural Networks like:

- **Long Short-Term Memory:** This type of network uses gates to decide what to remember and what to forget.

- **Gated Recurrent Unit:** This type of network is similar to Long Short-Term Memory but it is simpler and uses fewer parameters.

- **Bidirectional Recurrent Neural Network:** This type of network looks at the inputs in both the forward and backward directions, which helps it understand the context better.

These types of Recurrent Neural Networks can be used for different tasks that involve sequences of data. Recurrent Neural Networks are very useful for tasks that involve sequences, like speech recognition, text generation and machine translation. Recurrent Neural Networks are used in real-world applications and they are an important part of the field of artificial intelligence.

Q5(b). Illustrate the difference between SARSA and Q-Learning. [5 Marks]

SARSA and Q-Learning are two learning methods that help agents figure out the best way to do things by trying things out in an environment. They have some things in common. They are different in how they update what they think is the best action to take.

SARSA (State-Action-Reward-State-Action):

SARSA is a method that learns from what the agent's actually doing including when it tries new things.

Q-Learning:

Q-Learning is a method that learns what the best way to do things is, no matter what the agent is doing now.

For example:

Imagine a grid with a cliff. SARSA learns to take a path that is safe because it thinks about the chance that it might fall off the cliff when it tries new things. Q-Learning learns to take the path near the cliff because it always thinks the agent will make the best choice and it does not worry about the risk of trying new things.

Both SARSA and Q-Learning are methods, in learning and people choose one or the other based on how much risk they are willing to take and what they need to do. SARSA and Q-Learning are used in situations because SARSA and Q-Learning have different strengths.

Q6(a). Demonstrate the phases we need for doing Neural Network Analysis. [5 Marks]

Neural Network Analysis involves a series of steps each to building a good and reliable model. The key steps are:

Phase 1: Problem Definition and Data Collection

The first step is to define the problem. This could be a classification, regression or clustering task. Once defined we collect data from sources like databases, sensors, APIs or public datasets. The quality and quantity of data directly affect how well the neural network performs. We also need to understand the business context and define how we will measure success (using accuracy, precision, recall, F1-score, etc.).

Phase 2: Data Preprocessing

Raw data is rarely ready for modeling. This step involves:

- * Handling missing values by either replacing them or removing them.
- * Scaling features to a range (using Min-Max scaling or Z-score normalization) to make optimization easier.
- * Converting variables into numbers using one-hot encoding or label encoding.
- *. Extracting the most relevant features to reduce noise.
- * Splitting data into training, validation and test sets (70:15:15 or 80:10:10 splits).

Phase 3: Network Architecture Design

This step is about designing the network:

- * We choose the type of network (ANN, CNN, RNN, etc.) based on the problem.
- * We decide on the number of layers and how neurons each layer will have.
- * We select activation functions (like ReLU, sigmoid, tanh, softmax) for output layers.
- * We define the loss function (cross-entropy for classification, MSE for regression).
- * We choose the optimizer (SGD, Adam RMSprop). Set the learning rate.

Phase 4: Model Training

The network is trained on the training data by:

- * Making predictions.
- * Measuring the error.
- * Adjusting parameters to minimize the error.
- * Repeating this process across epochs monitoring for overfitting using the validation set.

Phase 5: Model Evaluation and Testing

The trained model is evaluated on the test set using metrics:

- * For classification: Accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC.
- * For regression: MAE, MSE, RMSE, R-squared.
- * This step checks if the model generalizes well.

Phase 6: Hyperparameter Tuning and Optimization

Based on evaluation results we fine-tune hyperparameters (like learning rate, batch size, number of layers, dropout rate) using techniques, like grid search, random search or Bayesian optimization to improve performance.

Phase 7: Deployment and Monitoring

The final model is deployed into production (via APIs, embedded systems or cloud services). After deployment we monitor the model to ensure it continues to perform with retraining scheduled as new data becomes available or performance degrades.

Q6(b). Reframe some algorithms commonly used with Image Recognition Systems. [5 Marks]

Image recognition systems use machine learning and deep learning algorithms to identify, classify and interpret visual content. The commonly used algorithms are:

1. Convolutional Neural Networks (CNNs):

CNNs are the foundation of image recognition. They use layers to extract features from images. These features include edges, textures and shapes.

- **LeNet-5** is one of the CNNs. It was designed for handwritten digit recognition.
- **AlexNet** won the 2012 ImageNet competition. This made learning popular for image recognition.
- **VGGNet** uses very deep architectures with small filters. It achieves accuracy.

2. ResNet (Residual Networks):

ResNet introduced connections. These connections allow gradients to flow directly through the network. This enables training of deep networks without degradation. ResNet accuracy on complex image classification benchmarks.

3. GoogLeNet / Inception:

The Inception architecture uses convolutions of different filter sizes. It captures features at scales simultaneously. This makes the network wider not deeper. It improves efficiency while maintaining high accuracy.

4. YOLO (You Only Look Once):

YOLO is a real-time object detection algorithm. It predicts bounding boxes and class probabilities. It is extremely fast. This makes it ideal for applications requiring real-time processing like driving and surveillance.

5. Support Vector Machines (SVM):

Before the deep learning era SVMs were widely used for image classification. SVMs find the hyperplane. This hyperplane maximises the margin between classes in dimensional feature space. They are still effective for datasets.

6. Transfer Learning Models:

-trained models like VGG, ResNet and EfficientNet are fine-tuned for specific image recognition tasks. Transfer learning reduces training time and data requirements. This makes it feasible to build image classifiers with limited domain-specific data.

7. R-CNN (Region-based CNN) Family:

R-CNN, Fast R-CNN and Faster R-CNN are designed for object detection and localisation. They combine region proposals with CNN-based feature extraction. This achieves accuracy in identifying and localising objects within images.

These algorithms power image recognition systems. They are used in healthcare for imaging. They are used in vehicles, facial recognition, satellite imagery analysis and retail, for visual search.